

LAW DOME SUMMIT, Antarctica

WMO: 898110

Lat: 66.730S

Lon: 112.835E

Elev: 1368

StdP: 85.93

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-40.4	-38.0	N/A	N/A	N/A	N/A	N/A	N/A	22.8	-23.3	20.6	-21.7	3.9	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	-5.3	N/A	-6.7	N/A	-7.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	3.4	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.8	15.2	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-20.1	-10.9	-13.9	-18.6	-22.0	-24.7	-25.3	-25.1	-25.3	-24.9	-22.2	-16.4	-11.9		
	DBStd	6.92	2.37	3.40	3.92	5.03	5.60	4.94	5.70	5.22	5.11	4.05	3.55	2.51		
	HDD10.0	10999	647	670	886	961	1076	1061	1089	1093	1048	997	792	680		
	HDD18.3	14041	905	903	1144	1211	1334	1311	1347	1352	1298	1256	1042	938		
	CDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH27.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	5.3	6.7	6.0	5.9	5.5	3.1	3.9	3.0	2.7	5.6	6.9	7.0	7.1		
(15) Precipitation	PrecAvg	433	35	36	41	35	31	38	42	42	34	30	35	34		
	PrecMax	538	56	61	67	62	50	66	76	73	62	49	58	53		
	PrecMin	348	19	5	23	15	7	14	20	13	13	10	10	21		
	PrecStd	52	9	14	11	13	11	13	16	15	11	9	12	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.9	-4.4	-8.9	-10.6	-10.8	-13.0	-12.0	-10.4	-12.3	-11.3	-6.1	-2.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	-5.3	-6.8	-10.8	-12.1	-12.4	-15.9	-14.3	-13.6	-14.2	-13.7	-8.6	-5.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	-6.6	-8.1	-12.0	-13.9	-15.6	-17.6	-15.9	-16.4	-16.0	-15.5	-10.1	-7.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	-7.6	-9.1	-13.0	-15.5	-17.4	-19.2	-17.6	-18.2	-17.8	-16.8	-11.5	-8.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.0	6.2	6.6	6.1	5.4	5.4	5.2	6.2	6.4	6.1	5.6		
		MCDBR	7.0	7.4	7.1	7.2	6.0	7.7	5.6	5.8	7.3	7.7	8.2	8.7		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.214	0.202	0.185	0.151	0.102	N/A	0.105	0.149	0.177	0.185	0.200	0.214		
	taud		2.237	2.260	2.270	2.163	1.904	N/A	1.896	2.134	2.246	2.268	2.250	2.220		
	Ebn at Noon		1059	1022	948	782	417	0	368	768	943	1038	1072	1072		
	Edh at Noon		118	99	73	42	12	0	11	41	73	97	115	124		
(44) All-Sky Solar Radiation	RadAvg		8.47	5.99	3.14	1.07	0.16	0.00	0.06	0.59	2.22	4.91	7.85	9.40		
	RadStd		0.16	0.15	0.10	0.04	0.01	0.00	0.00	0.03	0.07	0.11	0.21	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page